

CCA103 SAD

Use Cases

Chapter 3



Chapter 3: Outline

Part I
Use cases and
User Goals

Part II
Event
Decomposition

Part III
Use cases and
CRUD

Part IV
Use Case
Diagrams

Learning Objectives

1. Explain the purpose of use cases in the analysis phase of the SDLC.
2. Describe the various parts of a use case and the purpose of each part.
3. Describe how use cases contribute to the functional requirements.
4. Explain why identifying user stories and use cases is the key to defining functional requirements.
5. Describe the two techniques for identifying use cases.
6. Apply the event decomposition technique to identify use cases.
7. Describe the notation and purpose for the use case diagram.
8. Draw use case diagrams by actor and by subsystem.

Quick Recap

- A key part of defining a new system is understanding **user requirements**, which are the tasks users need to accomplish with the system.
- Use cases are used to express and describe these user requirements clearly.
- Use cases help analysts understand how users will interact with the system and what the **system must do for them (user)**.
- Process models are used to further clarify requirements by showing processes, activities, and how data move between them.
- Both use cases and process models are important tools in the analysis phase, especially during interviews and workshops, to identify and illustrate user and functional requirements.

User Stories

- A user story is usually a **short sentence** written in the everyday language of the end user.
- It explains what the **user does as part of their work or the goal** they want to achieve using the system.
- User stories are an important concept in Agile development because they emphasize simplicity and user value.
- They capture functional requirements quickly and less formally than traditional requirements models by focusing on who, what, and why.
- Users and stakeholders are the ones responsible for identifying and defining the user stories.



User Stories

- In meetings with stakeholders, analysts encourage participants to write out each user story on an index card or on a shared whiteboard app.
- The objective is to get a potential user to articulate what he or she wants to do with the new system.
- A standard template helps users think through what they want and why they want it.
- The standard template for a user story looks like this:

“As a <role played>, I want to <goal or desire> so that <reason or benefit>.”

User Stories

“As a <role played>, I want to <goal or desire> so that <reason or benefit>.”

For example,

A bank teller might be:

“As a teller, I want to make a deposit to quickly serve more customers.”

“As a teller, I want to balance the cash drawer to ensure there are no errors.”

As a customer of the bank using an ATM, some user stories might be:

“As a bank customer, I want to withdraw cash and feel confident the stack of cash I get is the correct amount.”

“As a bank customer, I want to deposit a check and feel confident the deposit is recorded correctly.”

User Stories

- A final part of a user story is the **acceptance criteria**. These indicate the features that must be present for the user to be satisfied with the resulting implementation.
- They focus on **functionality**, not on features or user-interface design.
- For example, the following are the acceptance criteria for the user story “bank teller making a deposit”:
 1. Customer lookup must be by name or by account number.
 2. It would be nice to display the photo and signature of the customer.
 3. Any check hold requirements must be indicated.
 4. Current balance and new balance must be displayed.

Sample User Story

User Story

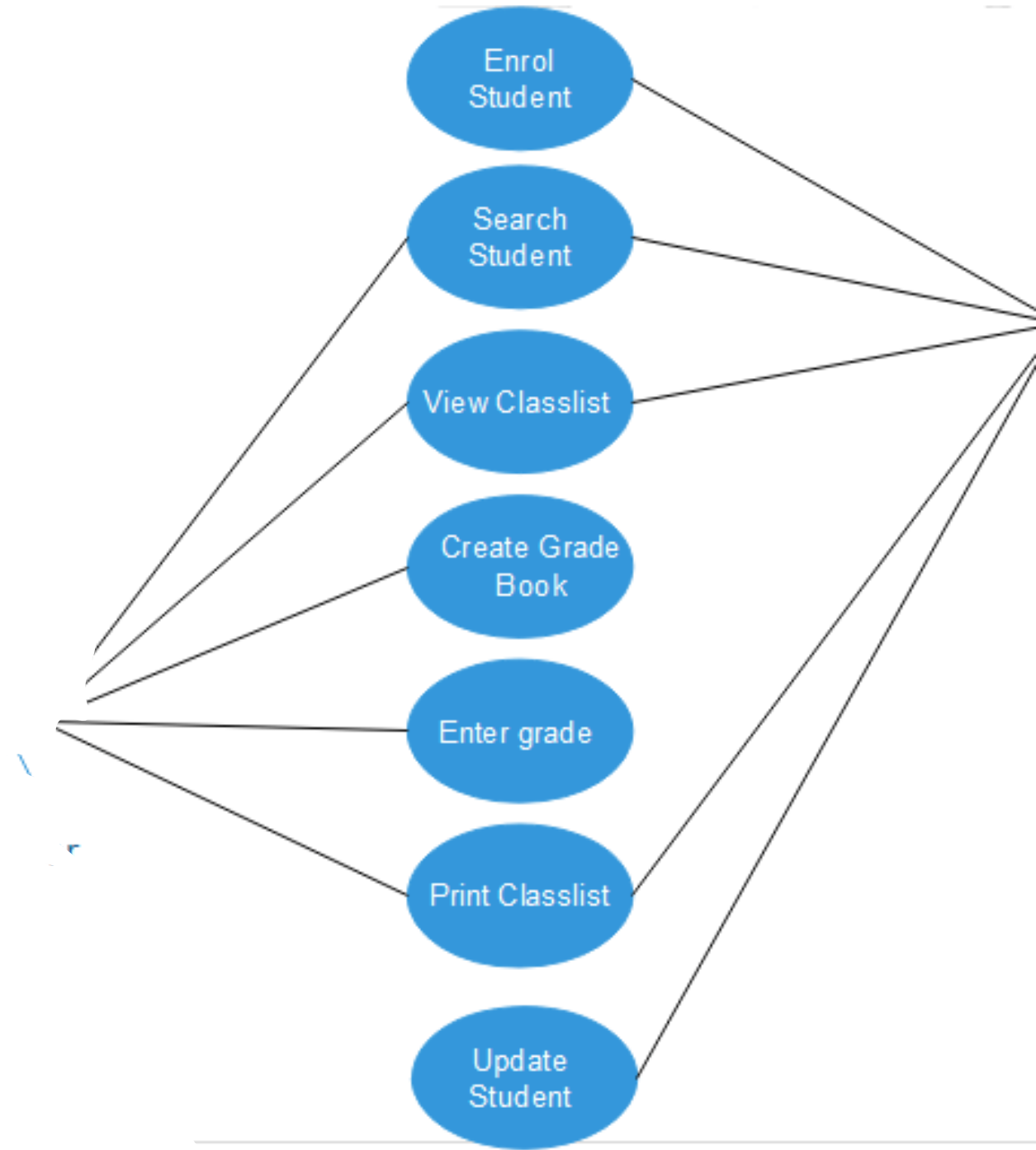
As a shipping clerk, I want to ship an order as accurately as possible as soon as the order details are available.

Acceptance Criteria:

- 1. Available order details must pop up on the screen when available.*
- 2. Portable display and scan device would cut time in half.*
- 3. Sort the items by bin location.*
- 4. Indicate number of items in stock for each item and mark backorder for those not available.*
- 5. Recommend shipper based on weight, size, and location.*
- 6. Print out shipping label for selected shipper.*

Use Cases

- Use case is an activity that the system performs, usually in response to a request by a user
- Use cases define **functional requirements**
- Analysts decompose the system into a set of use cases (functional decomposition)
- Two techniques for identifying use cases
 1. User goal technique
 2. Event decomposition technique
- Name each use case using *Verb-Noun*



User Goal Technique

- User Goal Technique is a technique to identify use cases by determining what specific goals or objectives must be completed by the system for the user
- This technique is the most common in the industry
- Simple and effective – is to ask users to describe their goals for using the new or updated system.
- Identify all the potential categories of users of the system
- Interview and ask them to describe the tasks the computer can help them with
- Probe further to refine the tasks into specific user goals, “I need to *Ship items, Track a shipment, Create a return.*”



User Goal Technique: Users & Goals

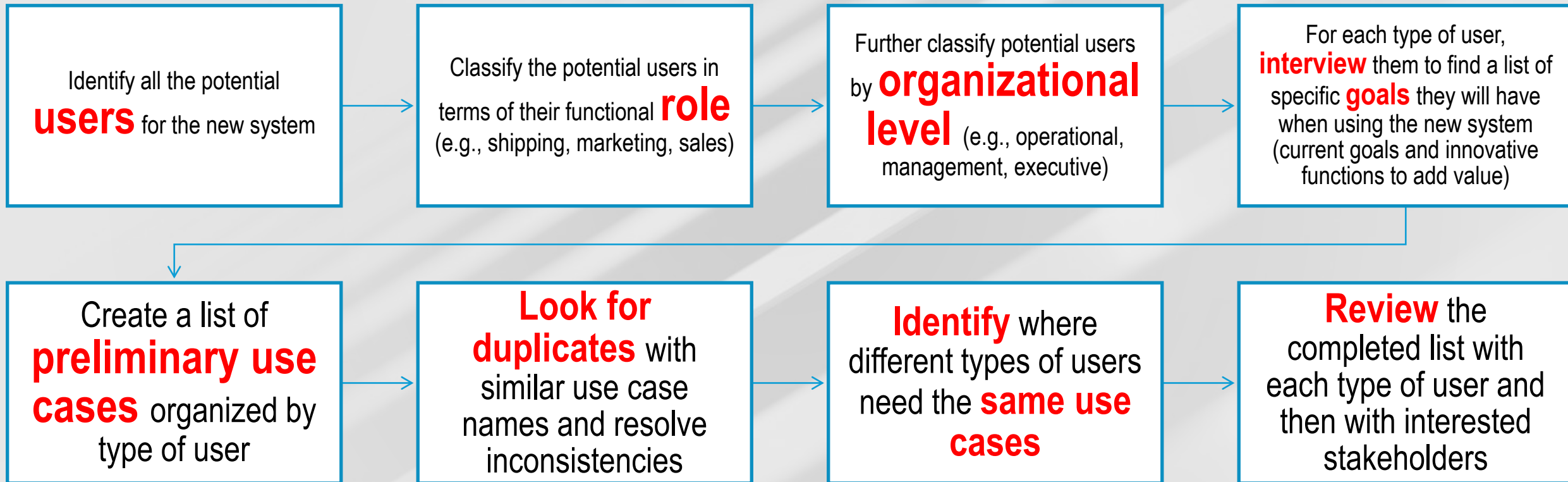
FIGURE 3-2 *Identifying use cases with the user goal technique*

User	User goal and resulting use case
Potential customer	Search for item Fill shopping cart View product rating and comments
Marketing manager	Add/update product information Add/update promotion Produce sales history report
Shipping personnel	Ship order Track shipment Create item return

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Source: Satzinger, J. W., Jackson, R. B., & Burd, S. D. (2015). Systems Analysis and Design in a Changing World (7th ed.). Course Technology CENGAGE Learning

User Goal Technique: Specific Steps



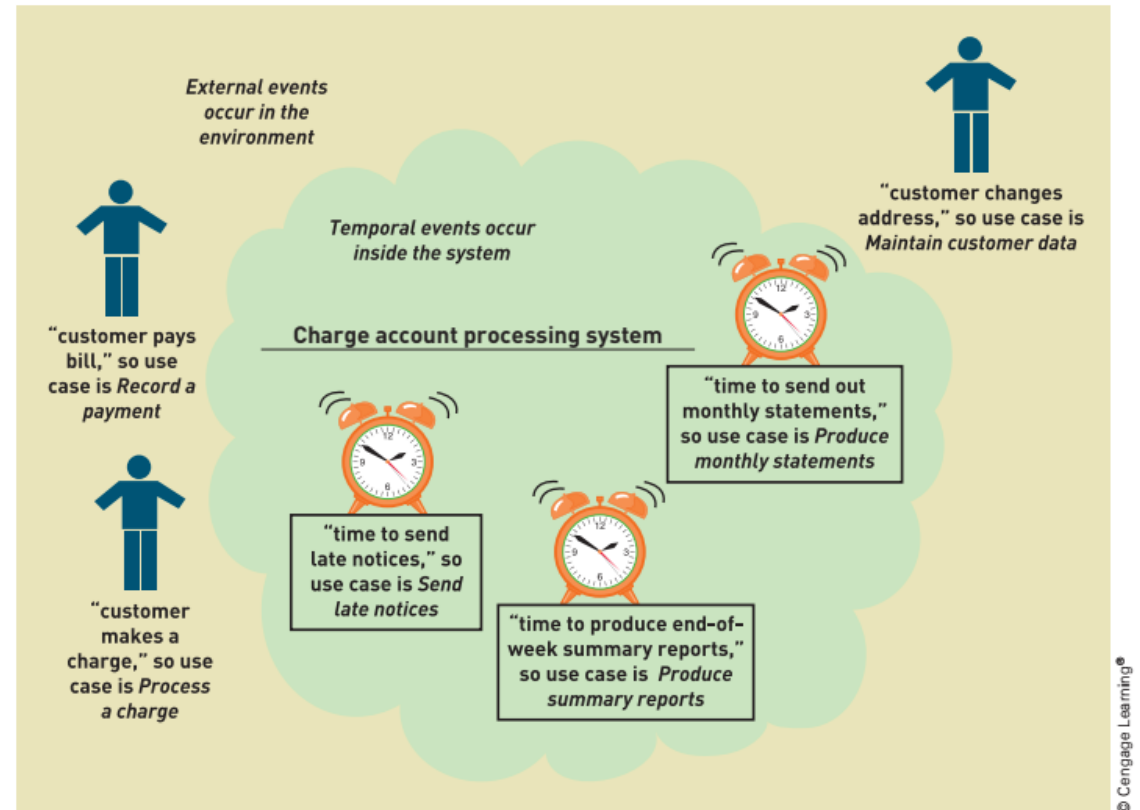
Event Decomposition Technique

- The event decomposition technique is the **most comprehensive** way to identify use cases.
- It starts by identifying all the **business events** that the information system must respond to.
- Each business event usually leads to one use case.
- This technique helps analysts define use cases at the correct level of detail.
- A good use case should represent a complete user goal—not too narrow and not too broad.

Events and Use Cases

1. A useful way to define system requirements is by asking what **business events** happen that require the system to respond.
2. This approach views the system as a **black box**, focusing only on inputs and outputs, not internal workings.
3. It helps analysts stay focused on the **overall scope of the system** and its interaction with external users or other systems.
4. In the retail charge account example, events triggered by customers include **paying a bill, making a charge, and changing an address**, each leading to a related use case.
5. Events can also be triggered by **time**, such as sending monthly statements, late notices, or weekly reports, which helps identify use cases at the **right level of analysis**.

FIGURE 3-3 Events in a charge account processing system that lead to use cases



Source: Satzinger, J. W., Jackson, R. B., & Burd, S. D. (2015). Systems Analysis and Design in a Changing World (7th ed.). Course Technology CENGAGE Learning

Types of Events

External Event

- an event that occurs outside the system, usually initiated by an external agent or actor

Temporal Event

- an event that occurs because of reaching a point in time

State Event

- an event that occurs when something happens inside the system that triggers some process
- reorder point is reached for inventory item

External Event Checklist

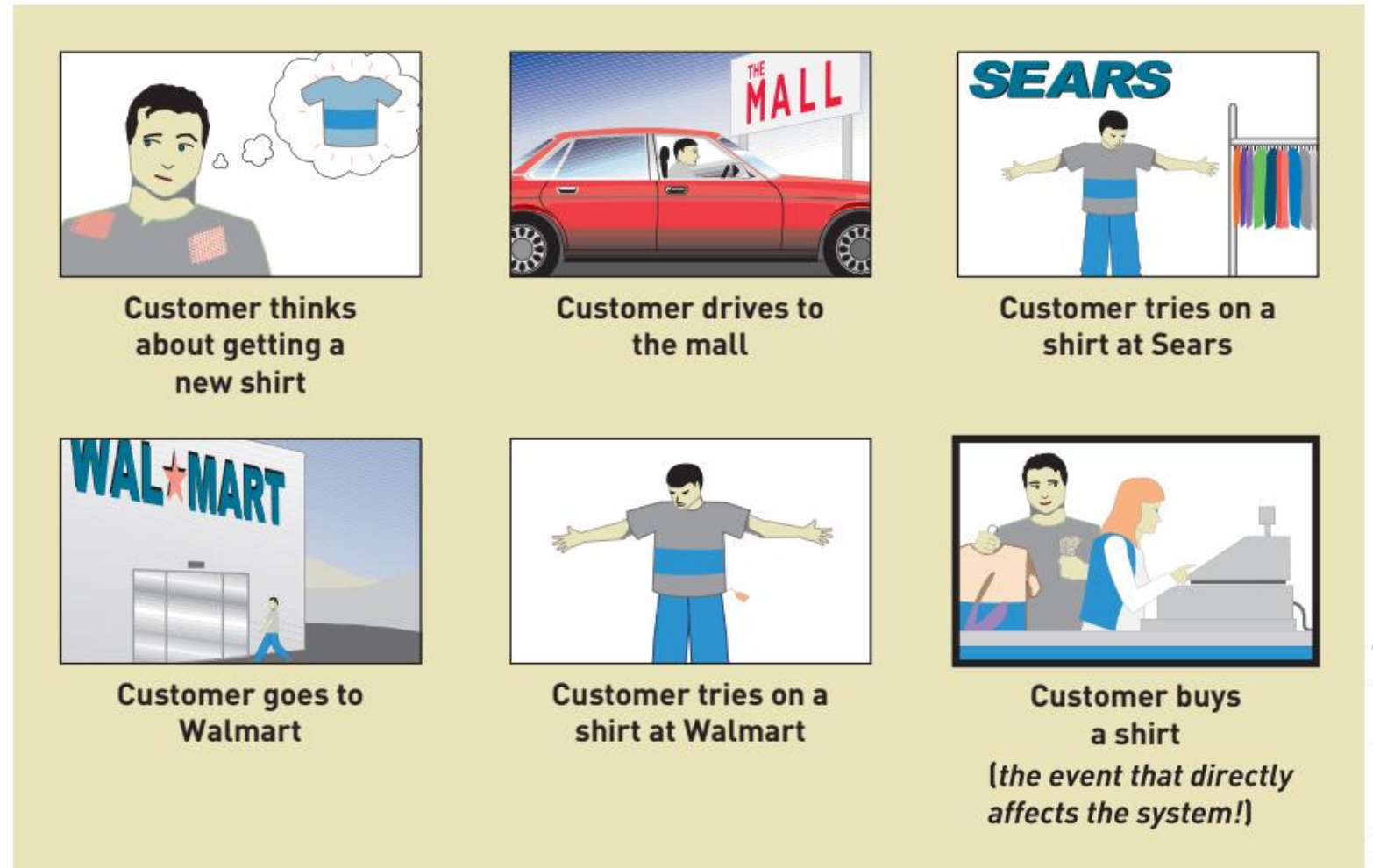
- **External agent or actor** wants something, resulting in a transaction
 - # Customer buys a product
- External agent or actor wants some information
 - # Customer wants to know product details
- External data changed and needs to be updated
 - # Customer has new address and phone
- Management wants some information
 - # Sales manager wants update on production plans

Temporal Event Checklist

- Internal outputs needed at **points in time**
 - # Management reports (summary or exception)
 - # Operational reports (detailed transactions)
 - # Internal statements and documents (including payroll)
- External outputs needed at points in time
 - # Statements, status reports, bills, reminders

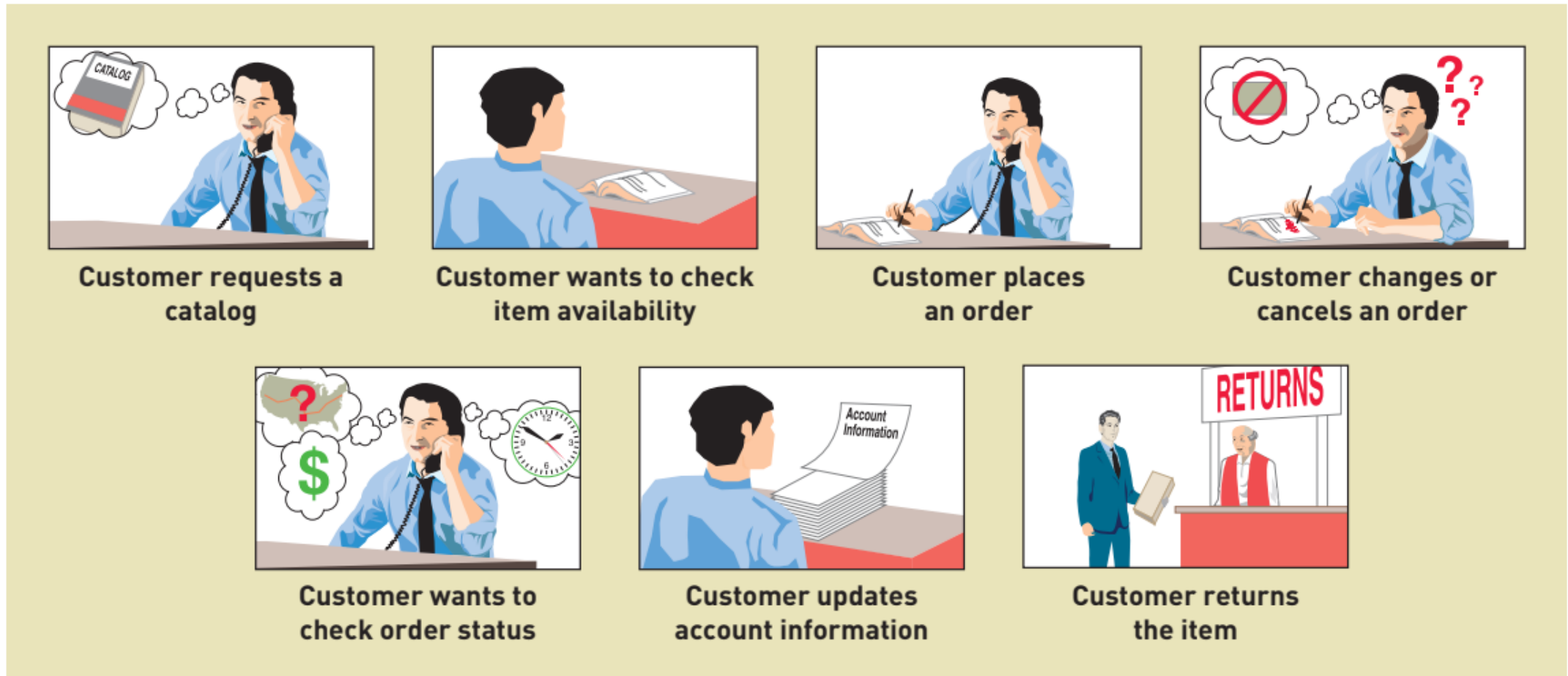
Finding the actual event that affects the system

FIGURE 3-6 Sequence of actions that lead up to only one event affecting the system



Tracing a sequence of transactions resulting in many events

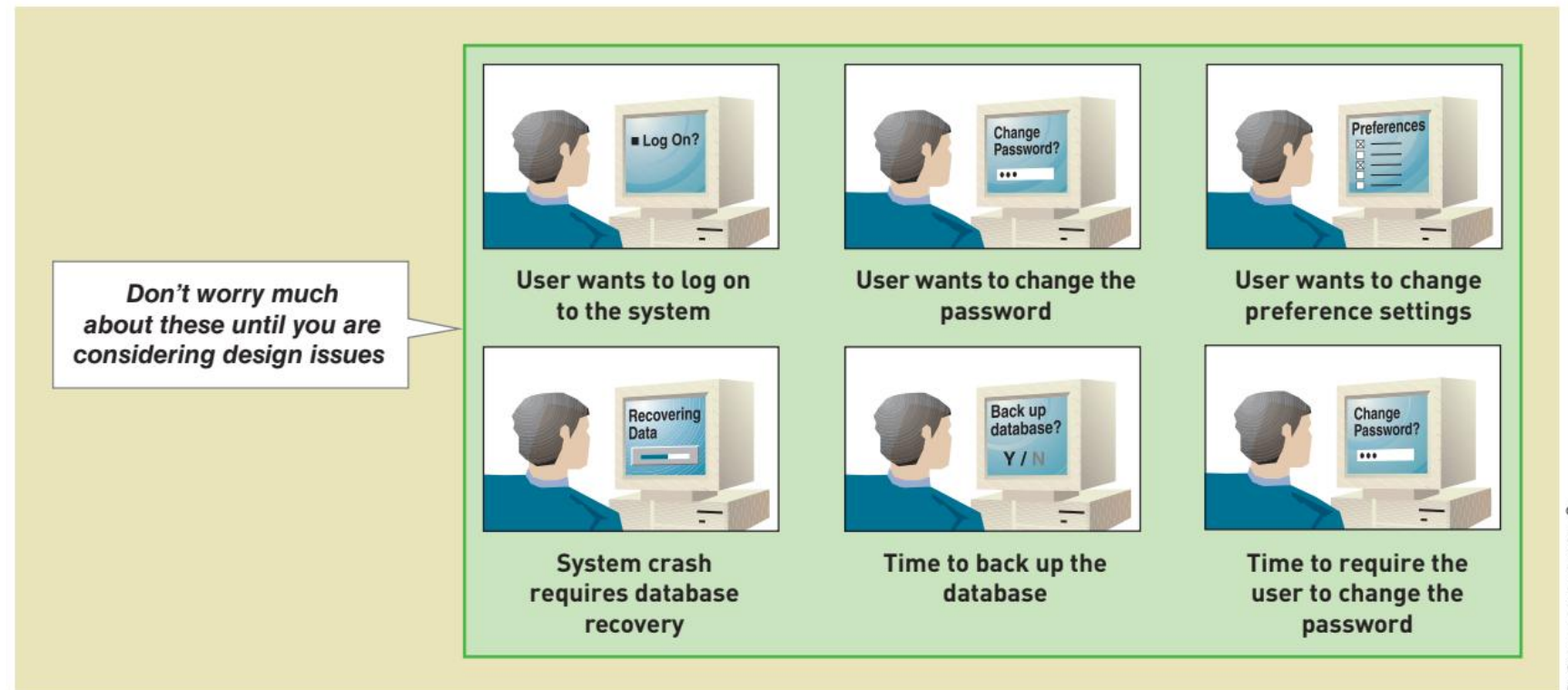
FIGURE 3-7 *The sequence of “transactions” for one specific customer resulting in many events*



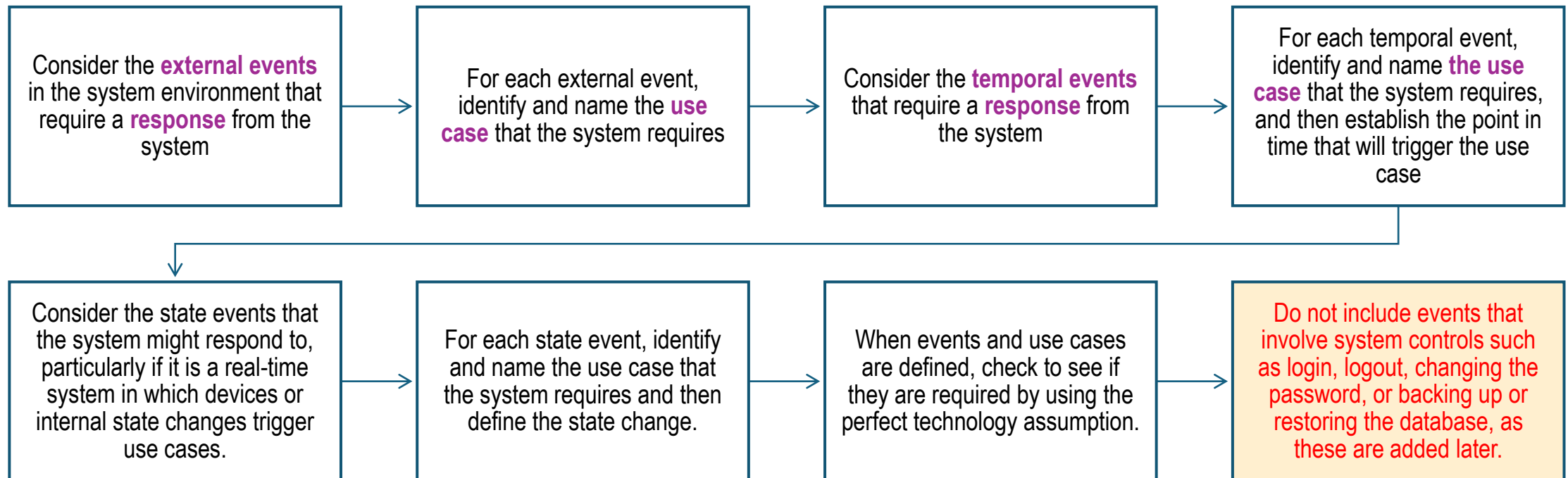
Perfect Technology Assumption

- Don't worry about functions built into the system due to technological and human limitations.
- Wait until the design.

FIGURE 3-8 Events deferred until designing system controls



Event Decomposition Technique: Specific Steps



Use Cases and Brief Use Case Descriptions

- A brief use case description is often a **one-sentence description** showing the main steps in a use case

FIGURE 3-9 *Use cases and brief descriptions*

Use case	Brief use case description
<i>Create customer account</i>	User/actor enters new customer account data, and the system assigns account number, creates a customer record, and creates an account record.
<i>Look up customer</i>	User/actor enters customer account number, and the system retrieves and displays customer and account data.
<i>Process account adjustment</i>	User/actor enters order number, and the system retrieves customer and order data; actor enters adjustment amount, and the system creates a transaction record for the adjustment.

Project Use Cases: Example 1

FIGURE 3-10a Use cases and actors for CSMS Sales Subsystem

CSMS Sales Subsystem	
Use cases	Users/actors
Search for item	Customer, customer service representative, store sales representative
View product comments and ratings	Customer, customer service representative, store sales representative
View accessory combinations	Customer, customer service representative, store sales representative
Fill shopping cart	Customer
Empty shopping cart	Customer
Check out shopping cart	Customer
Fill reserve cart	Customer
Empty reserve cart	Customer
Convert reserve cart	Customer
Create phone sale	Customer service representative
Create store sale	Store sales representative

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FIGURE 3-10b Use cases and actors for CSMS Order Fulfillment Subsystem

CSMS Order Fulfillment Subsystem	
Use cases	Users/actors
Ship items	Shipping
Manage shippers	Shipping
Create backorder	Shipping
Create item return	Shipping, customer
Look up order status	Shipping, customer, management
Track shipment	Shipping, customer, marketing
Rate and comment on product	Customer
Provide suggestion	Customer
Review suggestions	Management

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Project Use Cases: Example 2

FIGURE 3-10c Use cases and actors for CSMS Customer Account Subsystem

CSMS Customer Account Subsystem	
Use cases	Users/actors
Create/update customer account	Customer, customer service representative, store sales representative
Process account adjustment	Management
Send message	Customer
Browse messages	Customer
Request friend linkup	Customer
Reply to linkup request	Customer
Send/receive partner credits	Customer
View "mountain bucks"	Customer
Transfer "mountain bucks"	Customer

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FIGURE 3-10d Use cases and actors for CSMS Marketing Subsystem

CSMS Marketing Subsystem	
Use cases	Users/actors
Add/update product information	Merchandising, marketing
Add/update promotion	Marketing
Add/update accessory package	Merchandising
Add/update business partner link	Marketing

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FIGURE 3-10e Use cases and actors for CSMS Reporting Subsystem

CSMS Reporting Subsystem	
Use cases	Users/actors
Produce daily transaction summary report	Management
Produce sales history report	Management, marketing
Produce sales trends report	Marketing
Produce customer usage report	Marketing
Produce shipment history report	Management, shipping
Produce promotion impact report	Marketing
Produce promotional partner activity report	Management, marketing

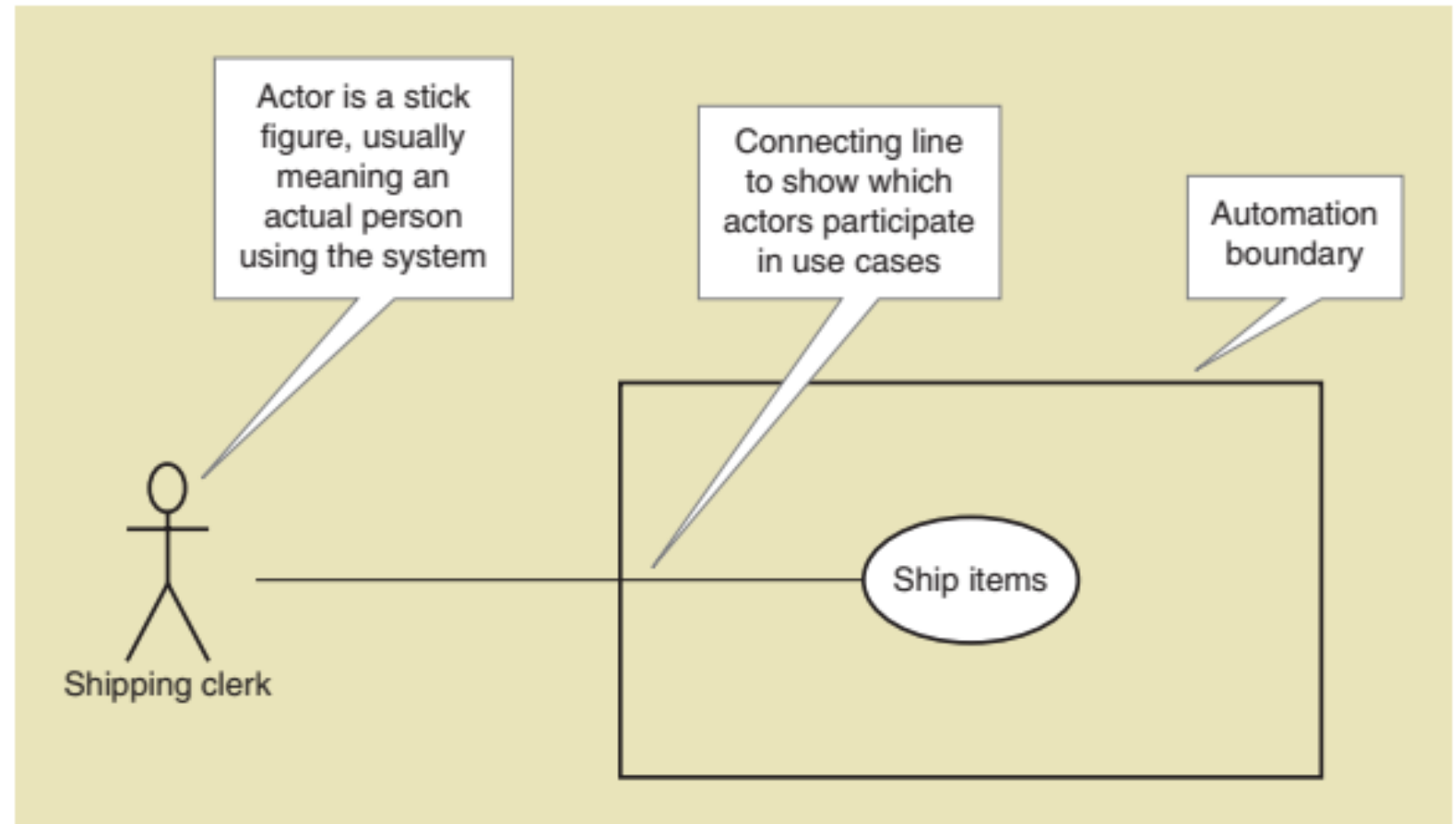
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Use Case Diagrams

- A use case diagram is a UML model used to graphically show use cases and their relationships to actors
- Recall UML is Unified Modelling Language, the standard for diagrams and terminology for developing information systems
- Actor is the UML name for an end user
- Automation boundary is the boundary between the computerized portion of the application and the users who operate the application

Use Case Diagrams Symbols

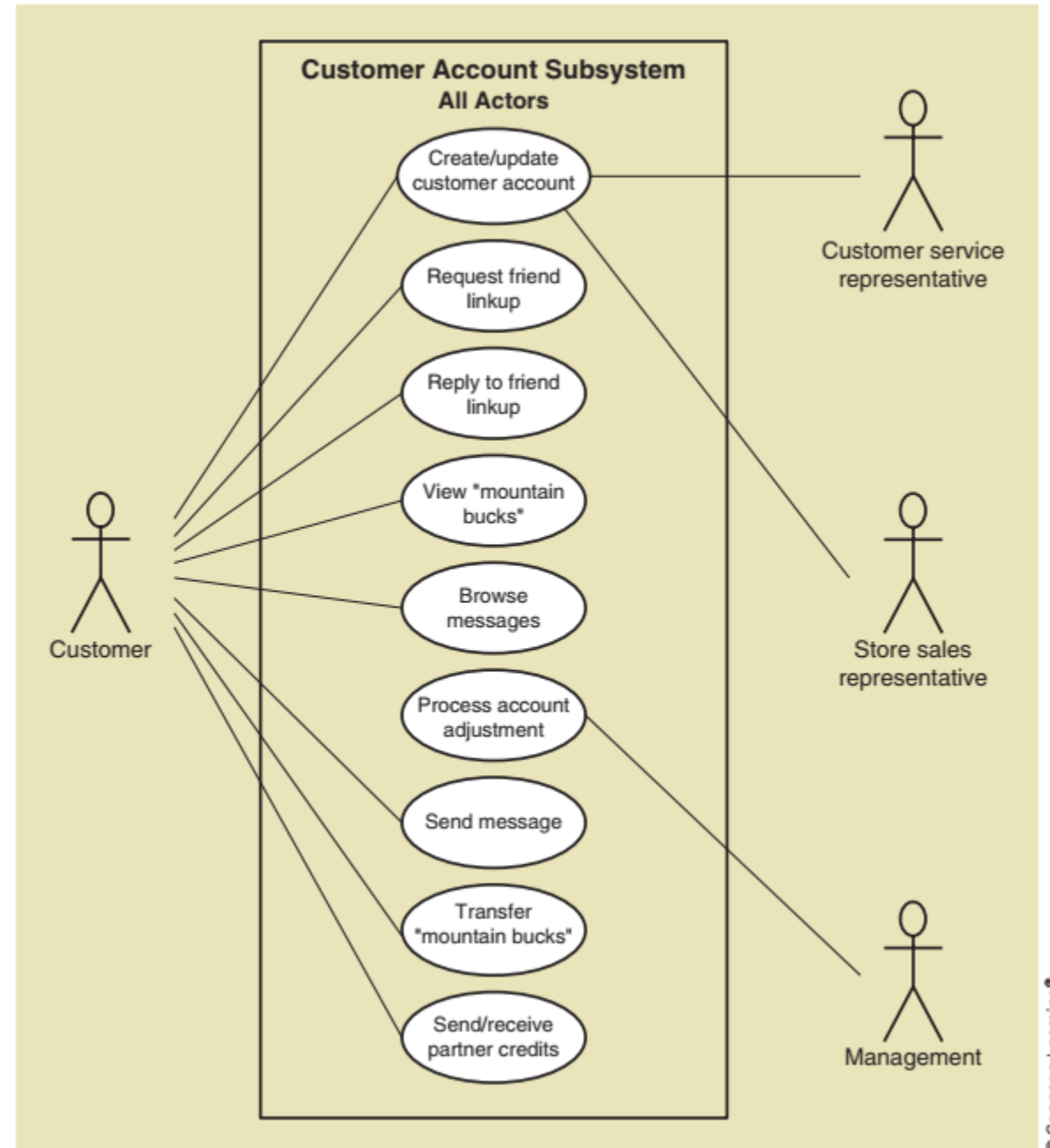
FIGURE 3-11 *A simple use case with an actor*



Use Case Diagrams

Draw for each subsystem

FIGURE 3-12 A use case diagram of the Customer Account subsystem for RMO, showing all actors



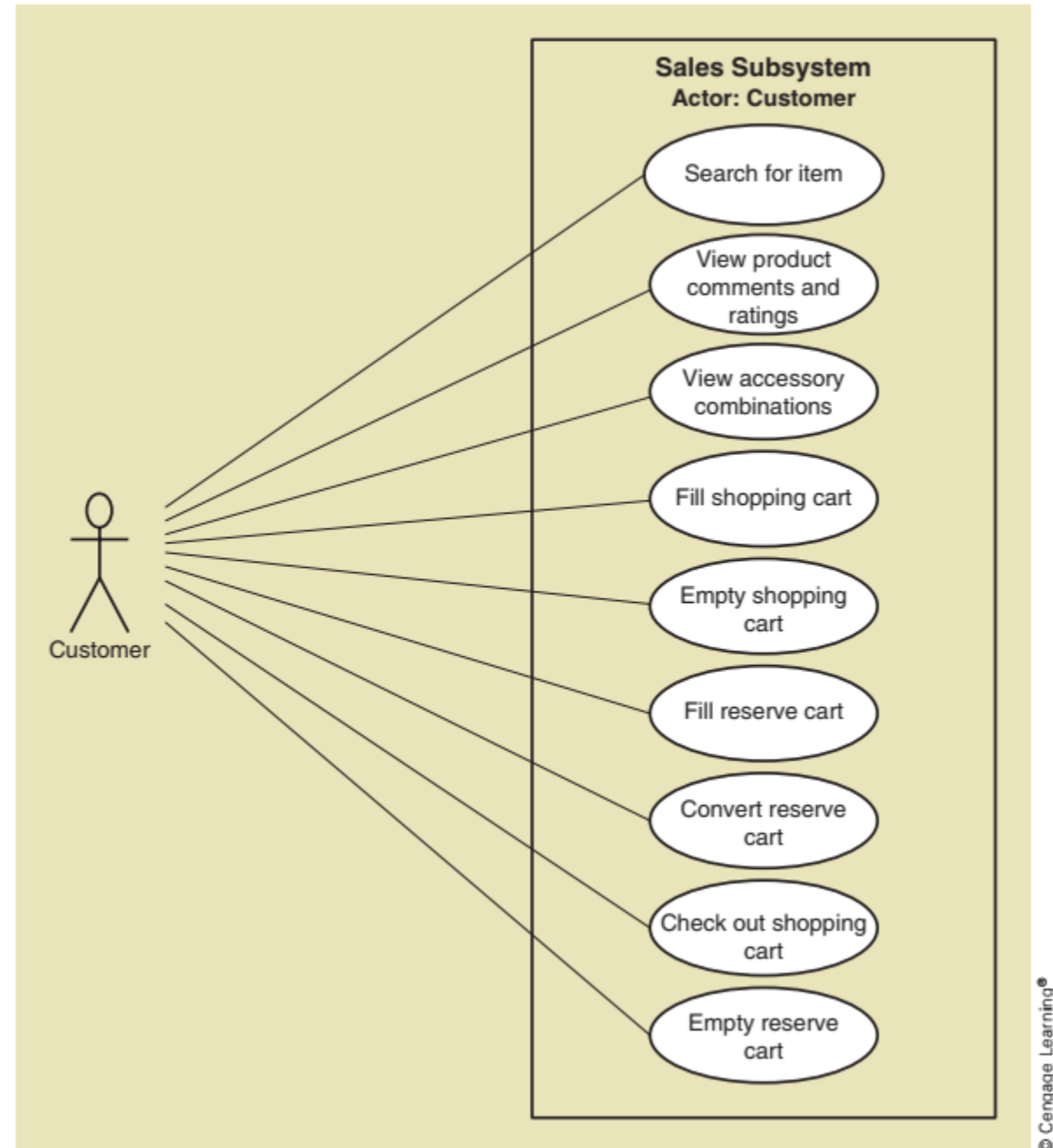
Use Case Diagrams

- There are many ways to organize use case diagrams for communicating with users, stakeholders, and project team members.
- One way is to show all use cases invoked by a particular actor (i.e., from the user's viewpoint).
- This approach is often used during requirements definition because the systems analyst may be working with a particular user and identifying all the functions that user performs with the system. Figure 3-13 illustrates this viewpoint, showing all the use cases involving the customer for the Sales subsystem.
- Figure 3-14 shows use cases involving the customer service representative and the store sales representative for the Sales subsystem.
- Analysts can expand this approach to include all the use cases invoked by any department, regardless of the subsystem, or all use cases important to a specific stakeholder.

FIGURE 3-13 All use cases involving the customer actor for the Sales subsystem

Use Case Diagrams

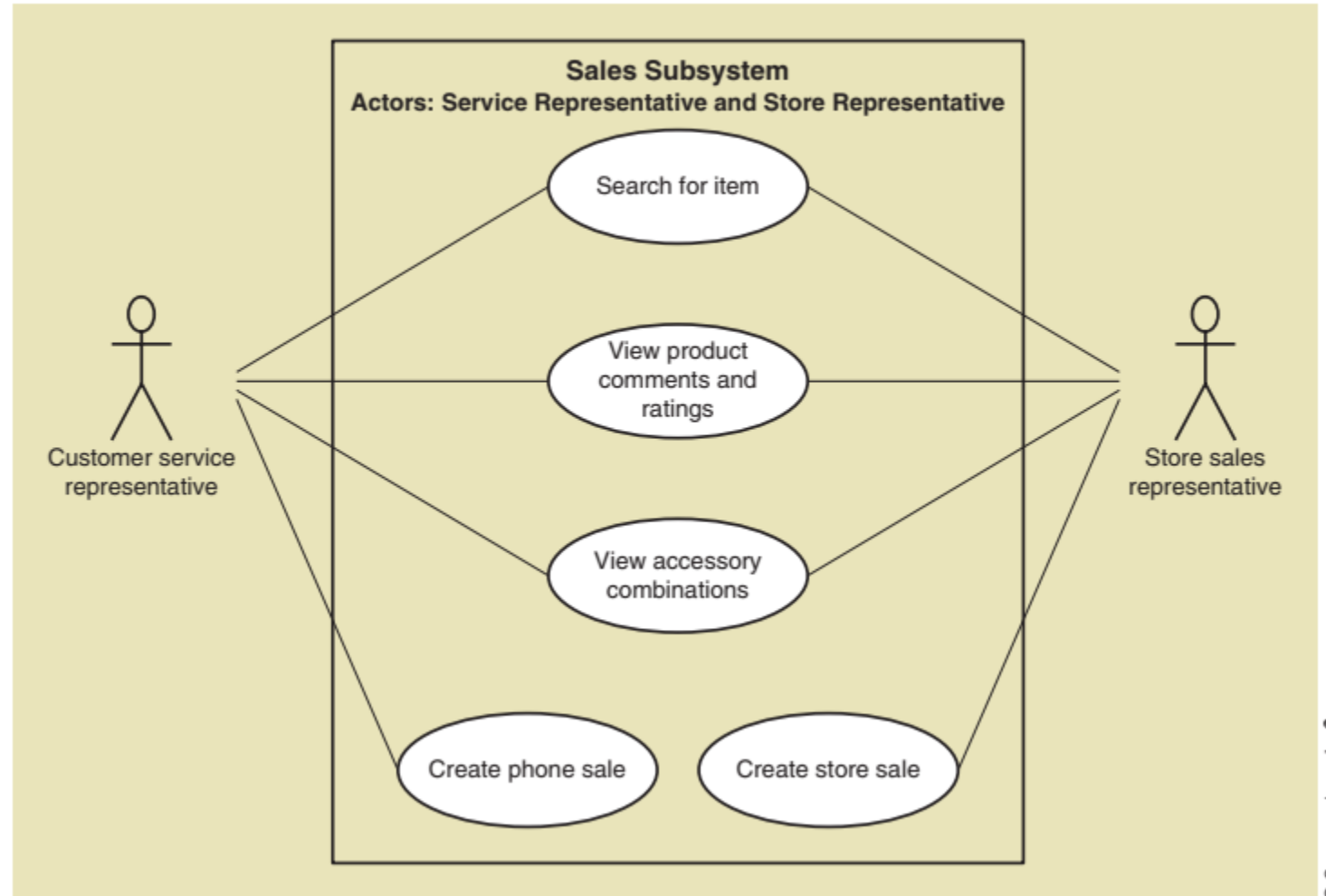
Draw for a single actor, such as a customer

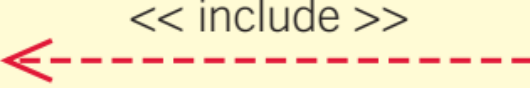


Use Case Diagrams

Draw for **internal actors**

FIGURE 3-14 Use cases involving the customer service representative and store sales representative for the Sales subsystem



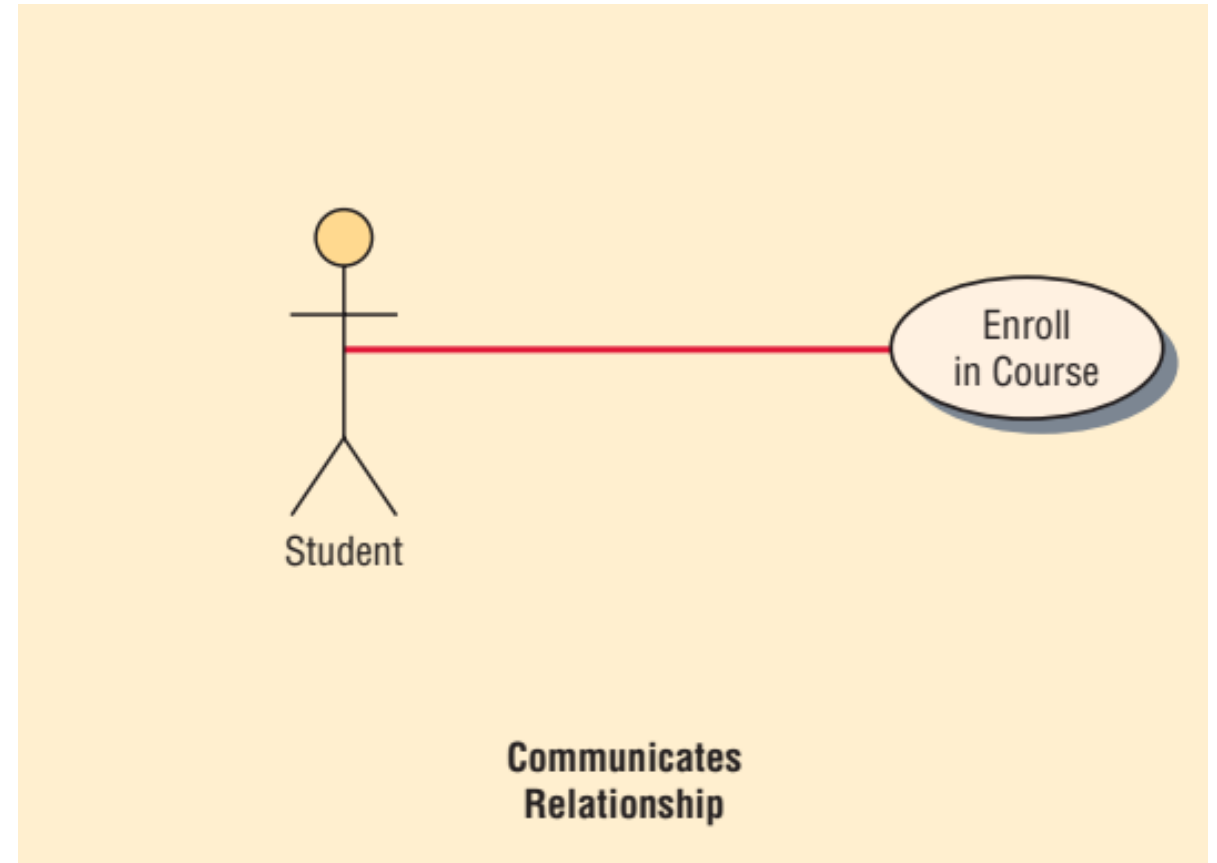
Relationship	Symbol	Meaning
Communicates		An actor is connected to a use case using a line with no arrowheads.
Includes		A use case contains a behavior that is common to more than one other use case. The arrow points to the common use case.
Extends		A different use case handles exceptions from the basic use case. The arrow points from the extended to the basic use case.
Generalizes		One UML “thing” is more general than another “thing.” The arrow points to the general “thing.”

Use Case Relationships

- Active relationships are referred to as behavioral relationships and are used primarily in use case diagrams.
- There are four basic types of behavioral relationships: communicates, includes, extends, and generalizes.
- Notice that all these terms are action verbs.

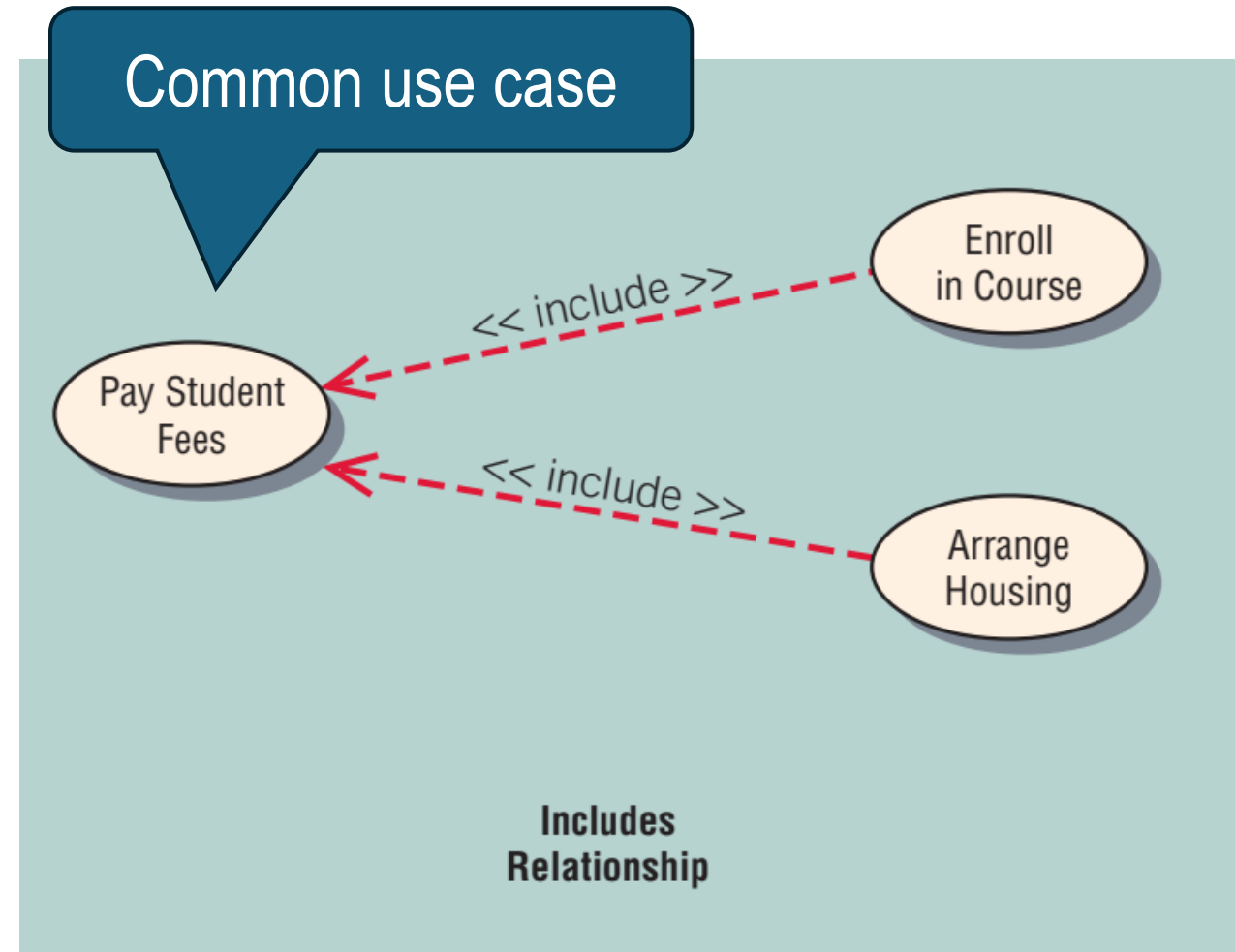
Use Case Relationships - Communicates

- The behavioral relationship communicates is used to connect an actor to a use case.
- Remember that the task of the use case is to give some sort of result that is beneficial to the actor in the system.
- Therefore, it is important to document these relationships between actors and use cases.



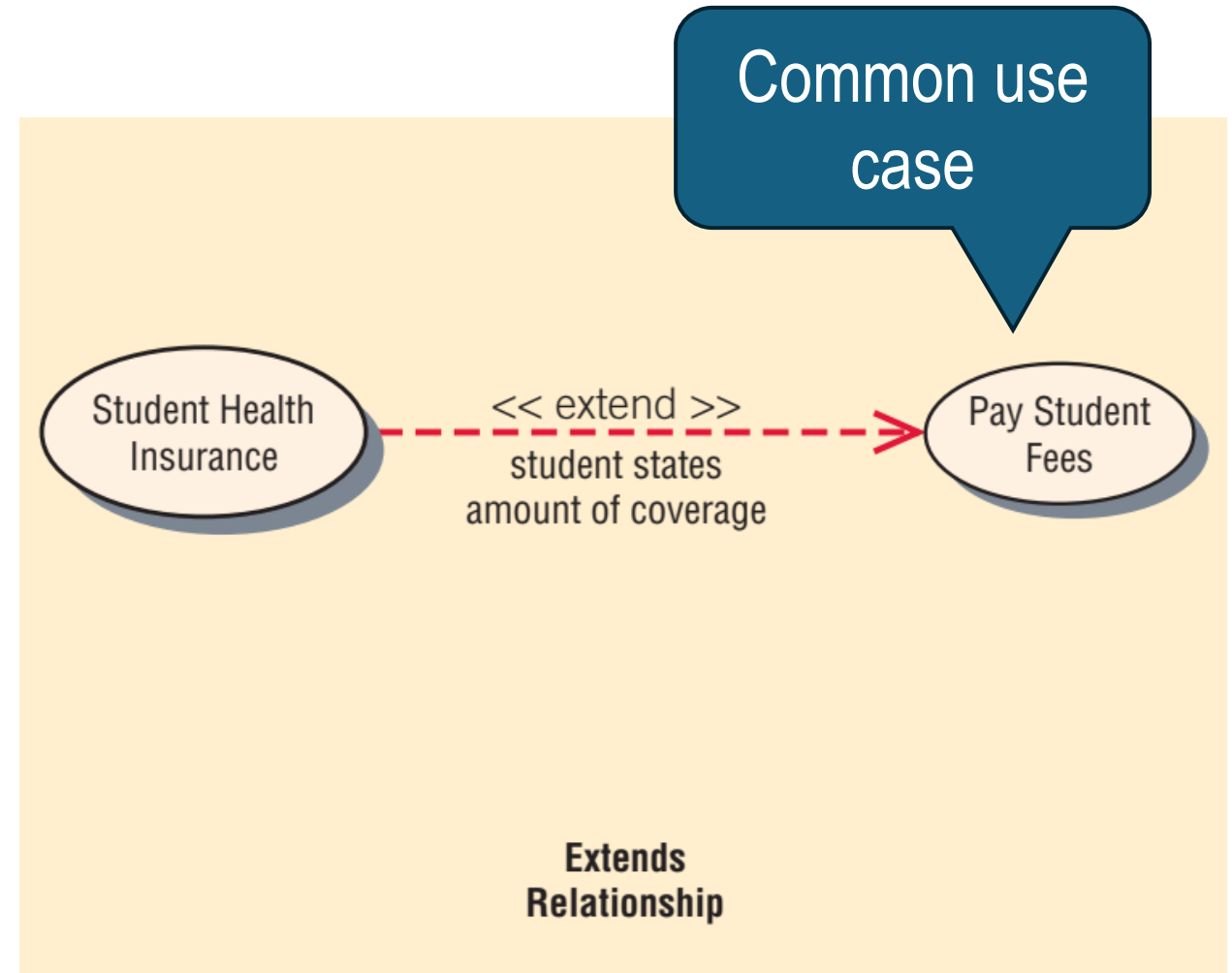
Use Case Relationships - Includes

- The includes relationship (also called uses relationship) describes the situation in which a **use case contains behavior that is common to more than one use case**.
- In other words, the common use case is included in the other use cases.
- A dotted arrow that points to the common use case indicates the includes relationship.
- An example would be a use case Pay Student Fees that is included in Enroll in Course and Arrange Housing, because in both cases students must pay their fees.
- This may be used by several use cases. The **arrow points toward the common use case**.



Use Case Relationships - Extends

- The extends relationship describes the situation in which **one use case possesses the behavior that allows the new use case to handle a variation or exception** from the basic use case.
- For example, the extended use case Student Health Insurance extends the basic use case PayStudent Fees.
- The **arrow goes from the extended to the basic use case.**



Use Case Relationships

– Includes vs Extends

Topping Select up to 2

Add Coffee Jelly (+RM1.90)

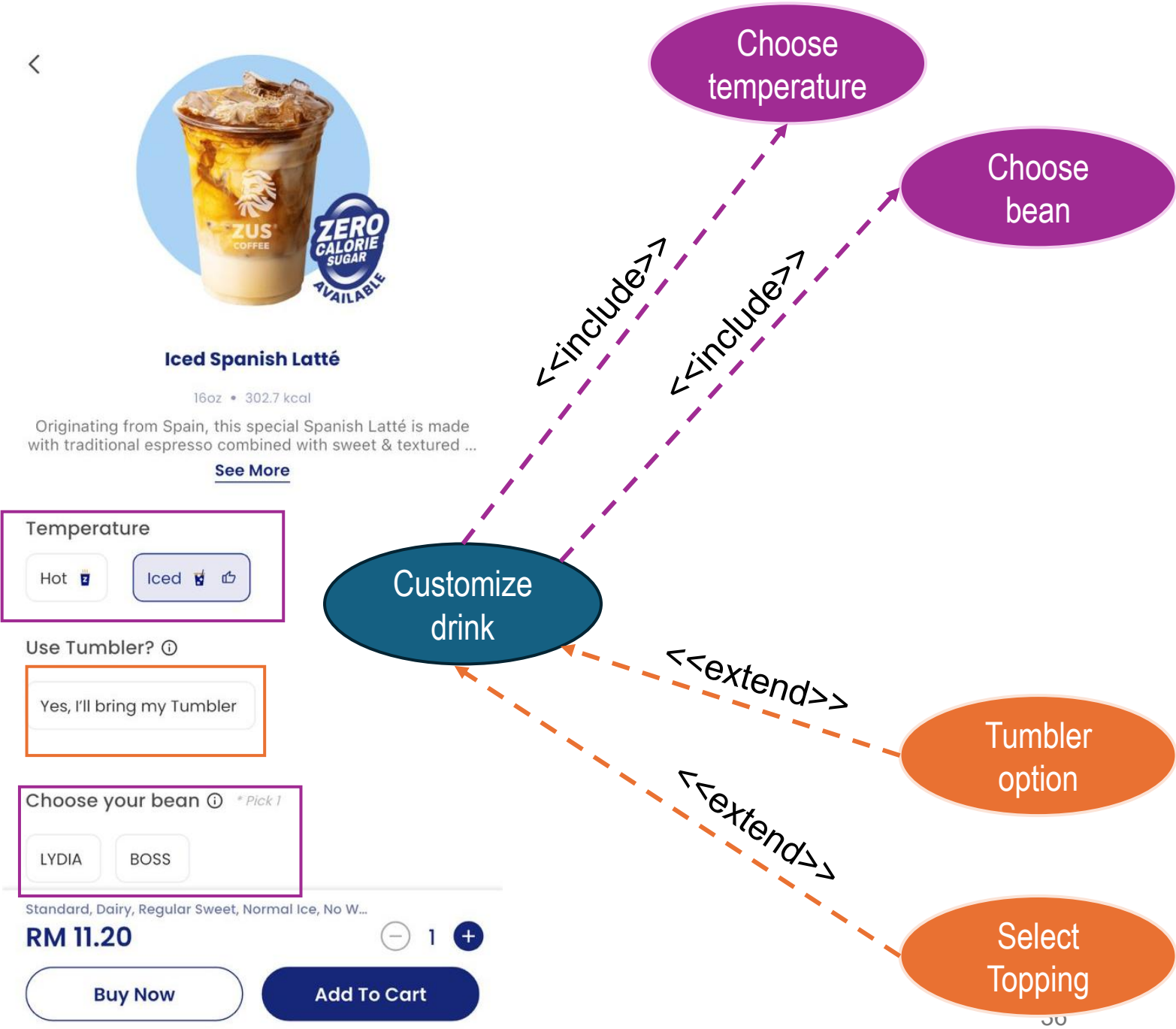
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Add Brownie Crumble (+RM0.60)

Mini Crystal Boba (+RM1.50)

Option ⓘ

Velvet Crème (+RM2)



Use Case Relationships – Includes vs Extends



1. Identify the main use case, and from the main use case, identify the include and extend relationship for the particular main use case
2. Based on the identified main use case, include and extend relationship, draw the use case diagram
3. Submit your answer via the e-learning portal

Use Case Relationships - Generalize

- The generalizes relationship implies that one thing is more typical than the other thing.
- This relationship may exist between two actors or two use cases.
- For example, a Part-Time Student generalizes a Student.
- Similarly, some of the university employees are professors.
- The arrow points to the general thing

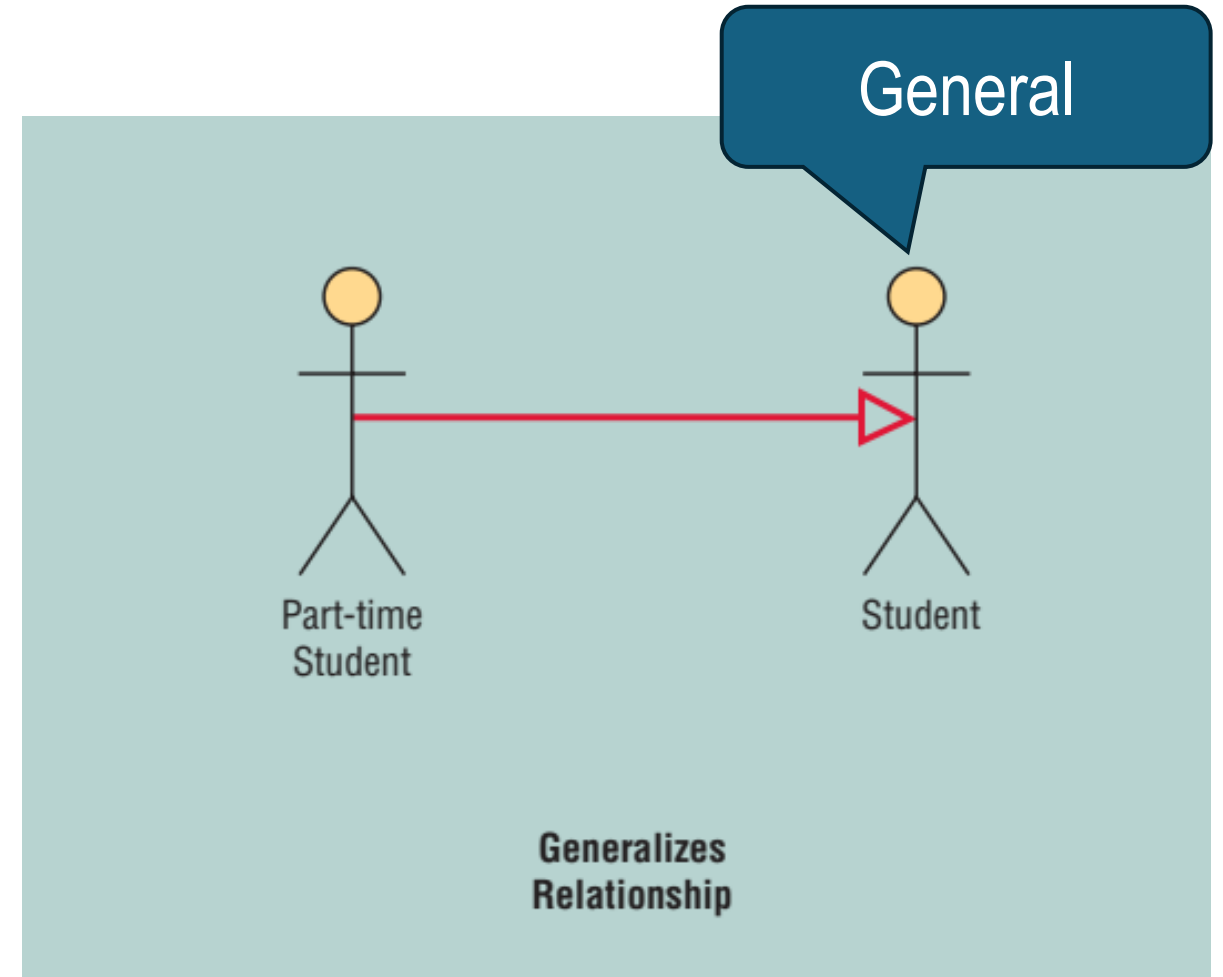


FIGURE 3-14 Use cases involving the customer service representative and store sales representative for the Sales subsystem

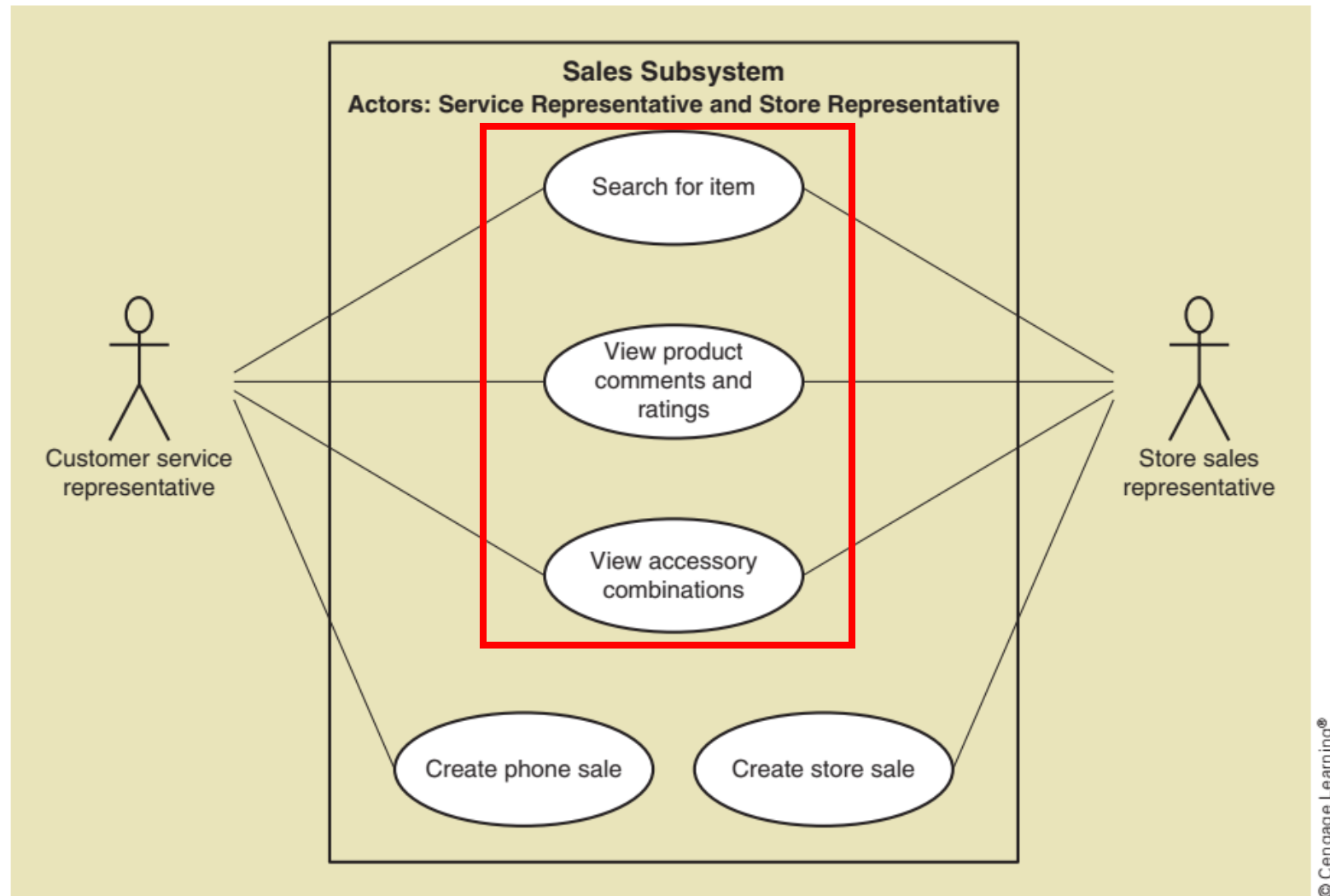
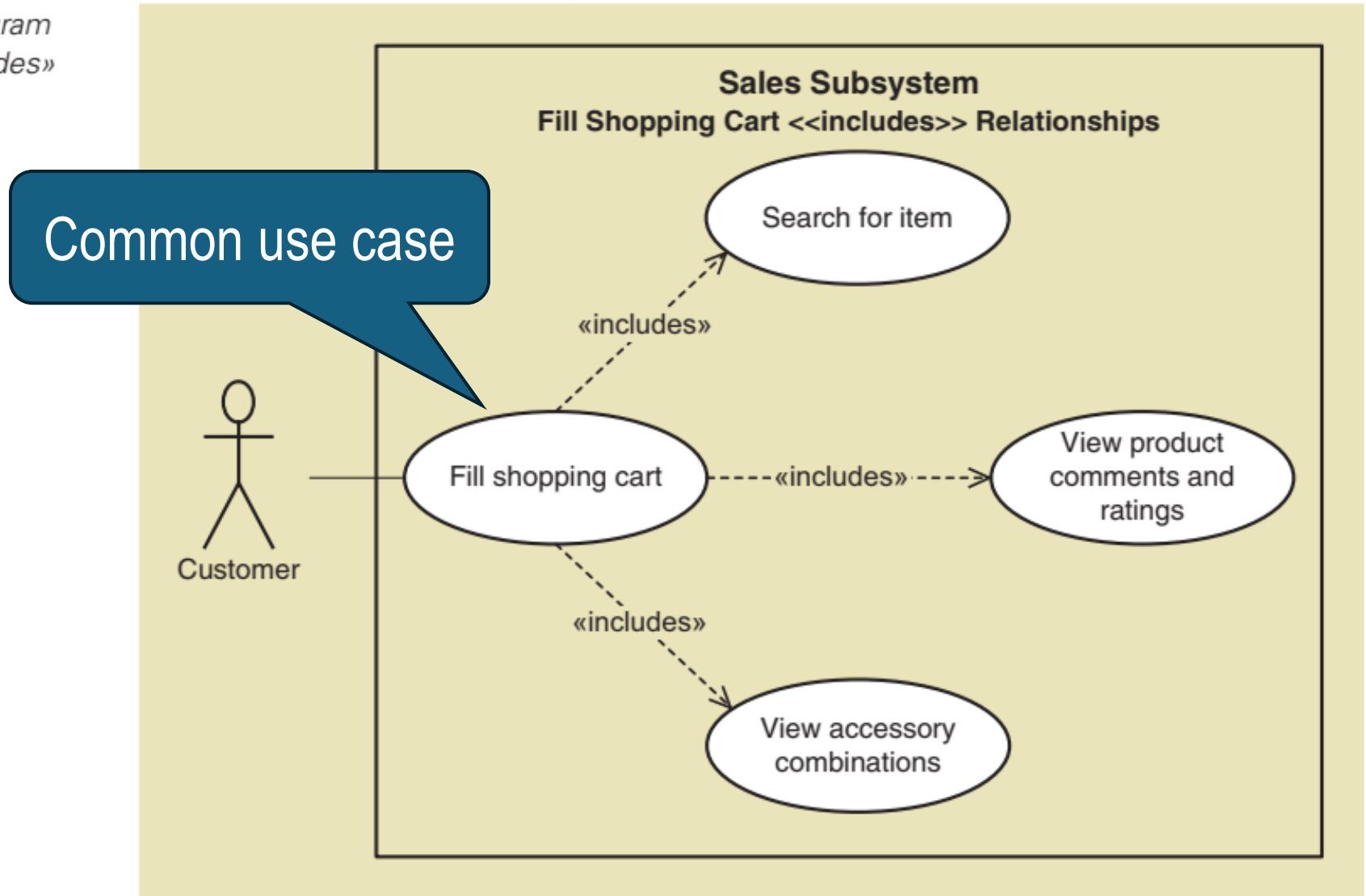


FIGURE 3-15 A use case diagram of the Fill shopping cart «includes» relationships



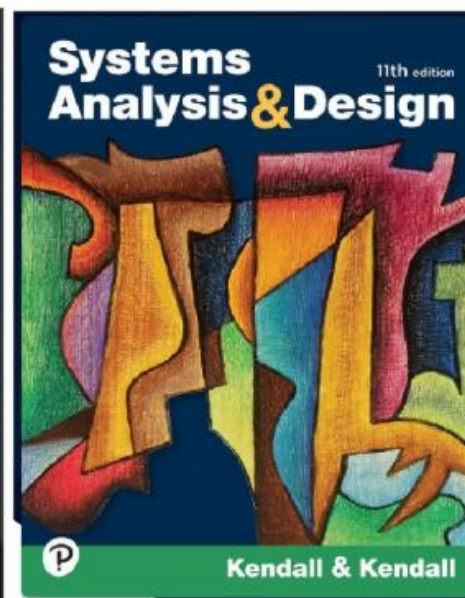
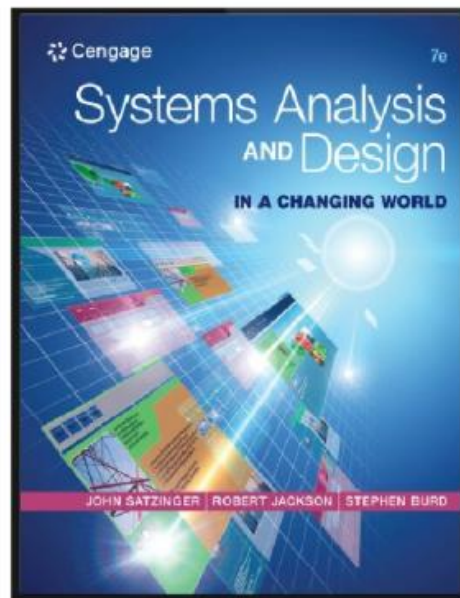
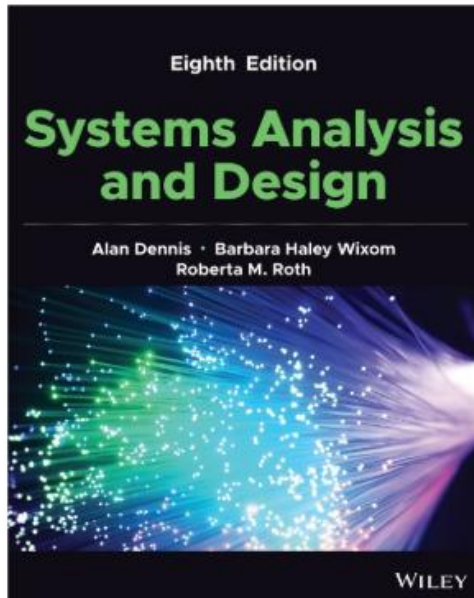
Use Case Diagrams: Steps

1. Identify all the stakeholders and users who would benefit by seeing a use case diagram
2. Determine what each stakeholder or user **needs** to review in a use case diagram: each **subsystem**, for each type of **user**, for use cases that are of interest
3. For each potential **communication** need, select the use cases and actors to show and **draw** the use case diagram.
4. There are many software packages that can be used to draw use case diagrams
5. Carefully name each use case diagram and then note how and when the diagram should be used to review use cases with stakeholders and users

Summary

- Use cases are the functions identified, the activities the system carries out, usually in response to a user request
- Two techniques for identifying use cases are the user goal technique and the event decomposition technique
- The user goal technique begins by identifying end users, called actors, and asking what specific goals they have when interacting with the system
- The event decomposition technique begins by identifying events that occur that require the system to respond.
- Three types of events include external, temporal, and state events
- Brief use case descriptions are written for use cases
- The use case diagram is the UML diagram used to show the use cases and the actors
- A variety of use case diagrams are drawn depending on the presentation needs of the analysis

References



- (R1) Dennis, A. Wixom B.H., Roth R. M. (2021). Systems Analysis and Design (8th ed.). John Wiley & Sons
- (R2) Satzinger, J. W., Jackson, R. B., & Burd, S. D. (2015). Systems Analysis and Design in a Changing World (7th ed.). Course Technology CENGAGE Learning
- (R3) Kendall, K. E., & Kendall, J. E. (2023). Systems Analysis and Design (11th ed.). Pearson